

WINNER ENGINEERING LEARNING HUB SERIES

FUNDAMENTAL TOPIC

AIR TERMINALS



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service

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- ▶ Sizing of Air Terminals
- ▶ Air Terminals Design Requirements

Introduction

Air Terminals are...

- ▶ Local air handling devices that manage and regulate input/output air in Duct System.

AIR TERMINALS



Linear Diffuser (2 Slot)



Round Supply air grille



Return air grille (Egg crate)



Bar grille



4-way or Square Supply
air diffuser



Double Deflection grille



Louver

INTRODUCTION

Definitions

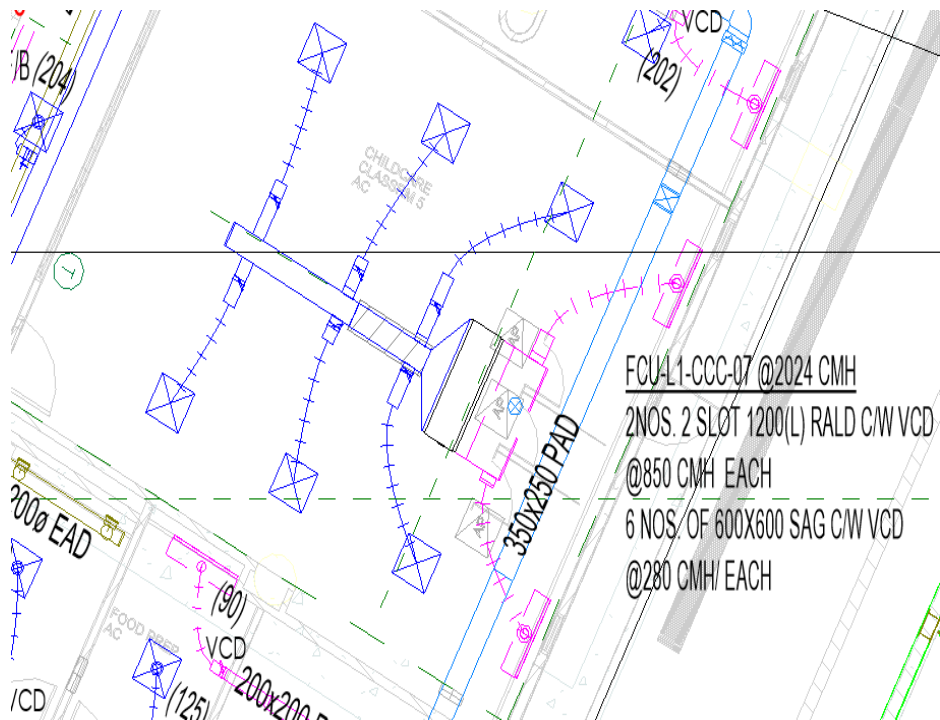
Outdoor Air Terminals

- ▶ Louvre: Allows Air to pass through while keeping out unwanted elements (e.g water, dirt , debri)

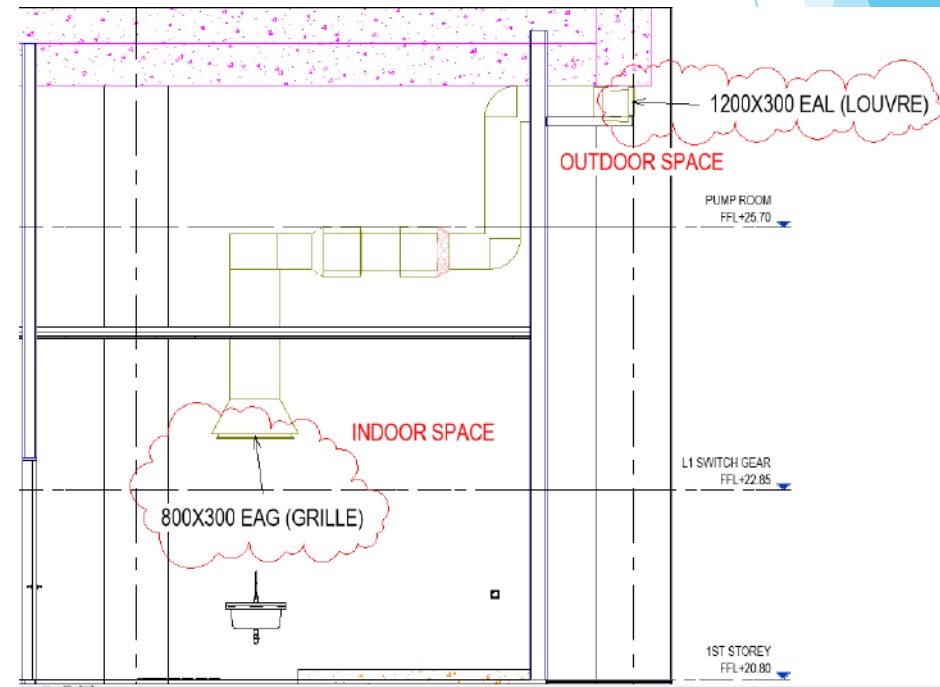
Indoor Air Terminals

- ▶ Diffusers & Grilles: Allow Supply and Exhaust of air into/out of an Indoor space
(Can come With/Without VCD)

Supply/Return Duct System (Plan View)



Kitchen Exhaust Duct System (Section View)



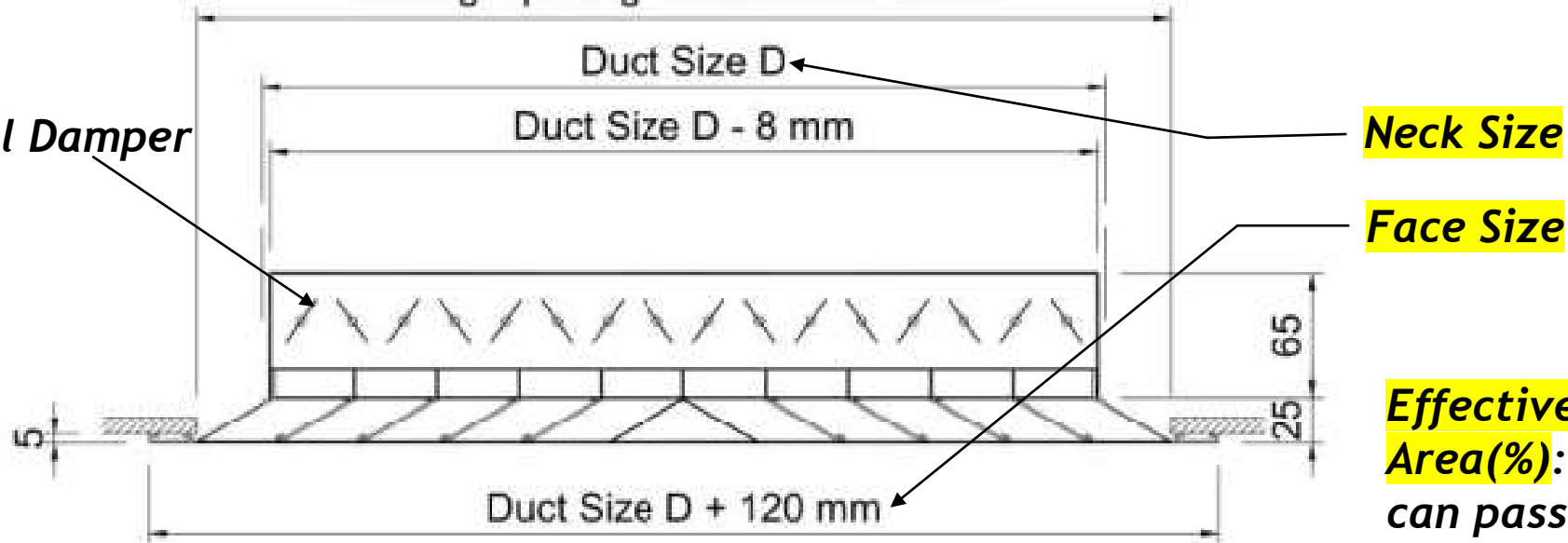
INTRODUCTION

Definitions

Section View of 4-Way Air Diffuser

(Surface Mount) Square or Rectangular Neck

Ceiling Opening = Duct Size + 70 mm



Neck Size

Face Size

Effective Free Opening Area(%): Area in which air can pass through

Throw Distance

(Only need to cover more than 50% of the room)

Face Velocity(m/s): Rate of air movement @face area

$$\rightarrow (\text{Air Flow(CMH)} \div 3600) \div \text{Face Area(m}^2\text{)}$$

Sizing of Air Terminals

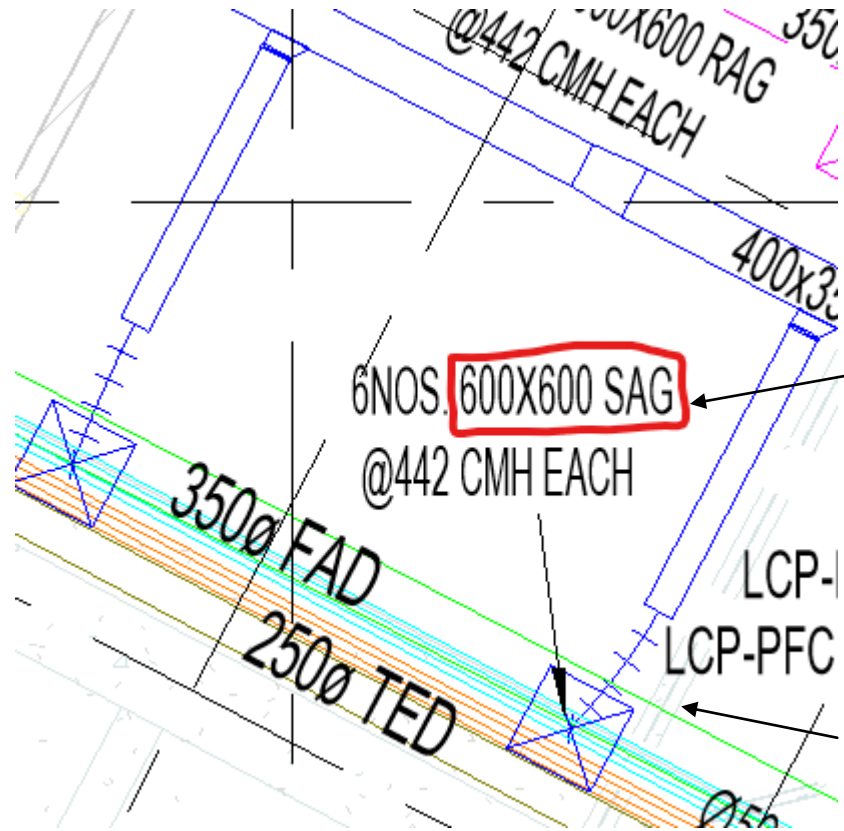
► Depends on...

1. Required Airflow (CMH)
2. Free Area(%)
3. Face Velocity(m/s)

► Important things to note

1. Free Area (70% for Grilles & Diffusers), (50% for Louvre)
2. Face Velocity (≤ 2.5 m/s)
 - ❖ High Face Velocity -> High Static Pressure -> High noise and Vibration level (**Undesirable**)
3. Use Standard Sizes (E.g 600x600)
 - ❖ Non-Standard size -> Customization work needed -> **More cost**

Sizing of Air Terminals



Face Size: Diffusers
Neck Size: Louvre and grilles

CEILING DIFFUSER

600x600 FACE SIZE

SIZE (mm)	CAP l/s	50	60	70	80	90	100	120	140	160	180	200	250	300	350	400	450	500	600	700
150X150	S.P.	11	15	20	26	32	39	58	78											
	NR	14	18	22	27	30	35	40	45											
	THROW	2	3	3	3.5	4	4.5	5	5.5											
200X200	S.P.			11	14	18	22	33	45	54	73									
	NR			15	18	21	25	30	35	40	45									
	THROW			3	3.5	3.5	4	4.5	5	5.5	6									
250X250	S.P.						9	15	18	20	31	40	52	71						
	NR						15	18	21	27	31	33	40	48						
	THROW						3.5	4	4.5	5	5.5	5.5	6.5	7						
300X300	S.P.						7	9	12	17	20	27	31	54	64	90				
	NR						12	15	18	20	23	28	35	40	45	50				
	THROW						3.5	4	4.5	5	5	6	6.5	6.6	7.5	7.5				
350X350	S.P.							5	7	8	10	15	21	30	36	70	55			
	NR							11	13	15	17	21	27	31	35	39	43			
	THROW							4	4	4.5	5	5.5	6	6.5	7	7	7.5			
400X400	S.P.								6	7	8	12	17	22	28	38	43	60		
	NR								11	13	15	18	25	27	31	35	38	44		
	THROW								4.5	4.5	5	5.5	6	6.5	7	7	7.5	8		
450X450	S.P.											6	10	13	18	23	32	35	49	65
	NR											13	17	19	24	29	32	35	40	45
	THROW											5	6	6	6.5	6.5	7	7	8	8

Sizing of Air Terminals

Winner Detail and Standards (Neck Size Requirement)

FLEXIBLE DUCT SIZING

AIR QUANTITY (CMH)	DIAMETER (mm)
0 – 250	150
251 – 450	200
451 – 750	250
751 – 1050	300
1050 – 1350	350
1351 – 1800	400

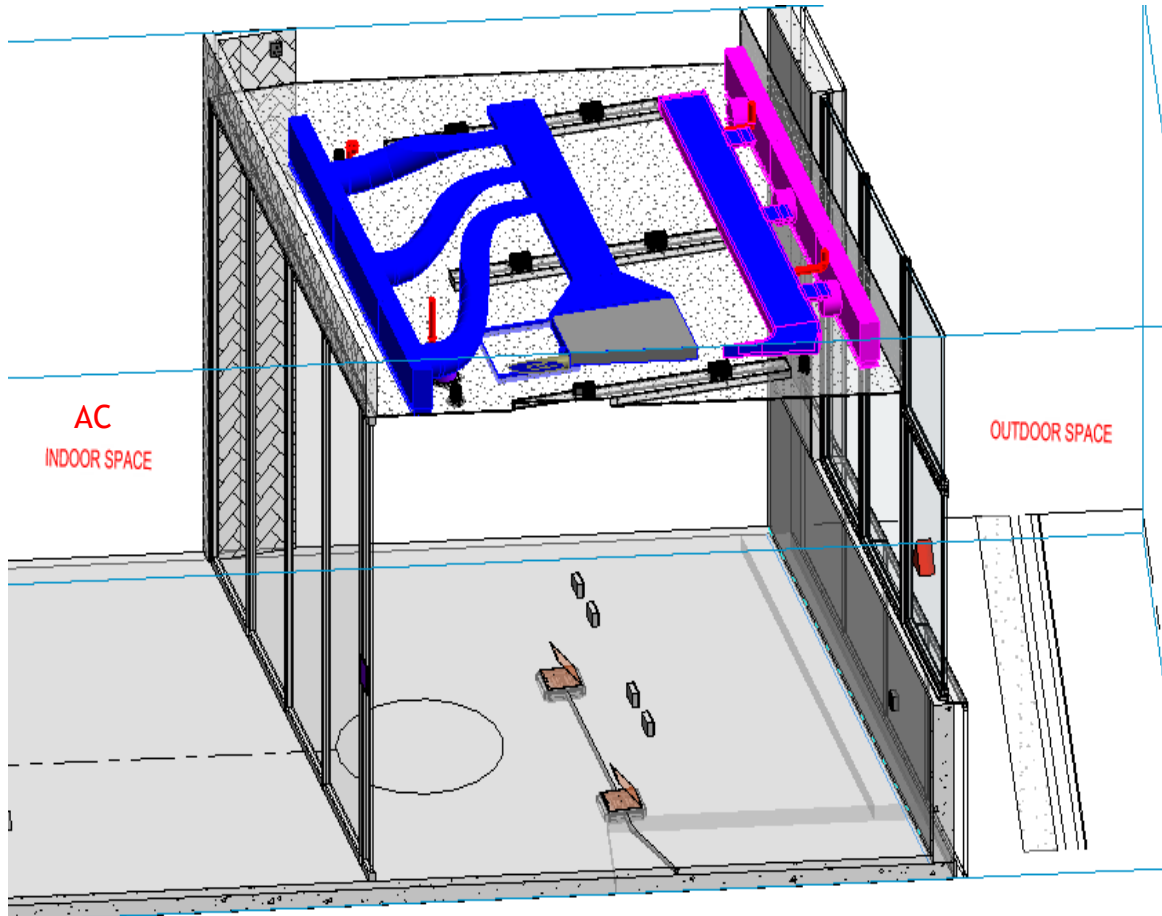
All flexible ducts shall not exceed 2m in length for all connection

SCHEDULE OF NECK SIZE FOR GRILLE AND DIFFUSER

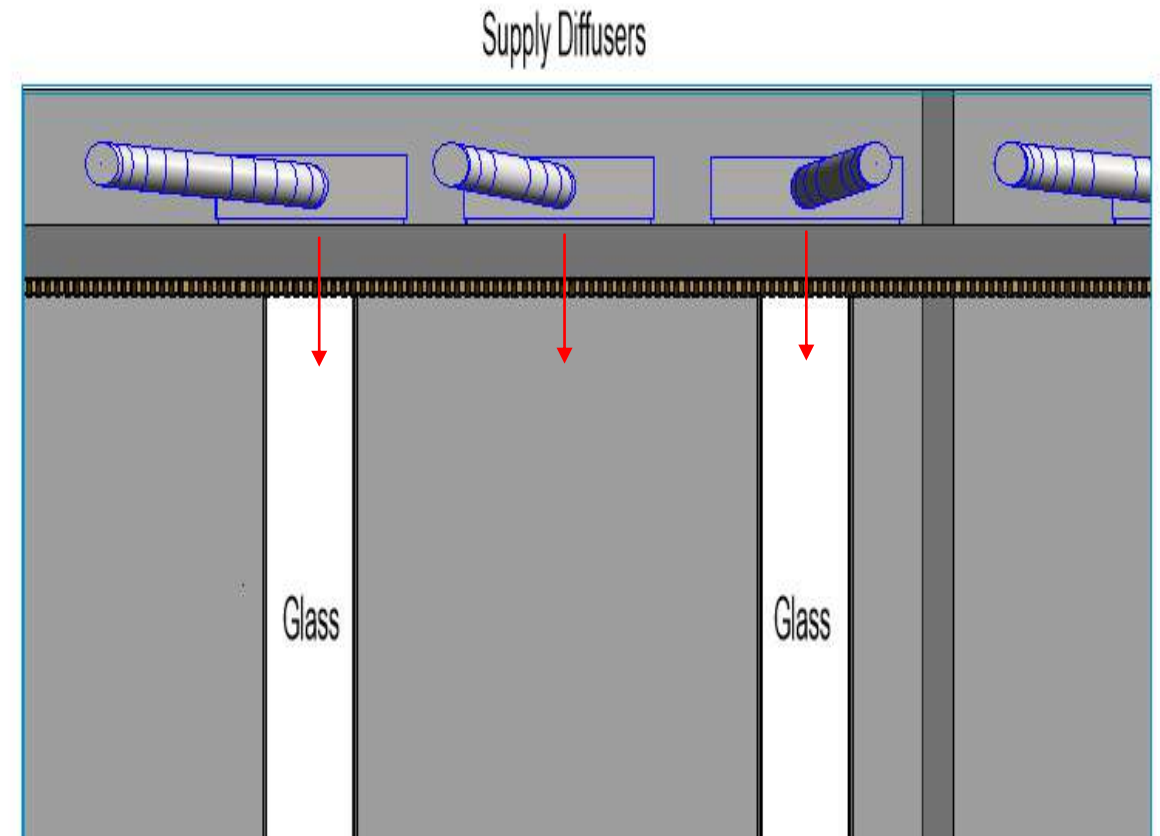
AIR QUANTITY (CMH)	SIZE (MMXMM)
0-68	100X100
69-162	150X150
163-280	200X200
281-442	250X250
443-638	300X300
639-867	350X350
868-1130	400X400
1131-1435	450X450
1436-1768	500X500
1769-2250	550X550

Design Requirements of Air Terminals

3D View of Supply/Return System



Section View of Supply Linear Diffusers

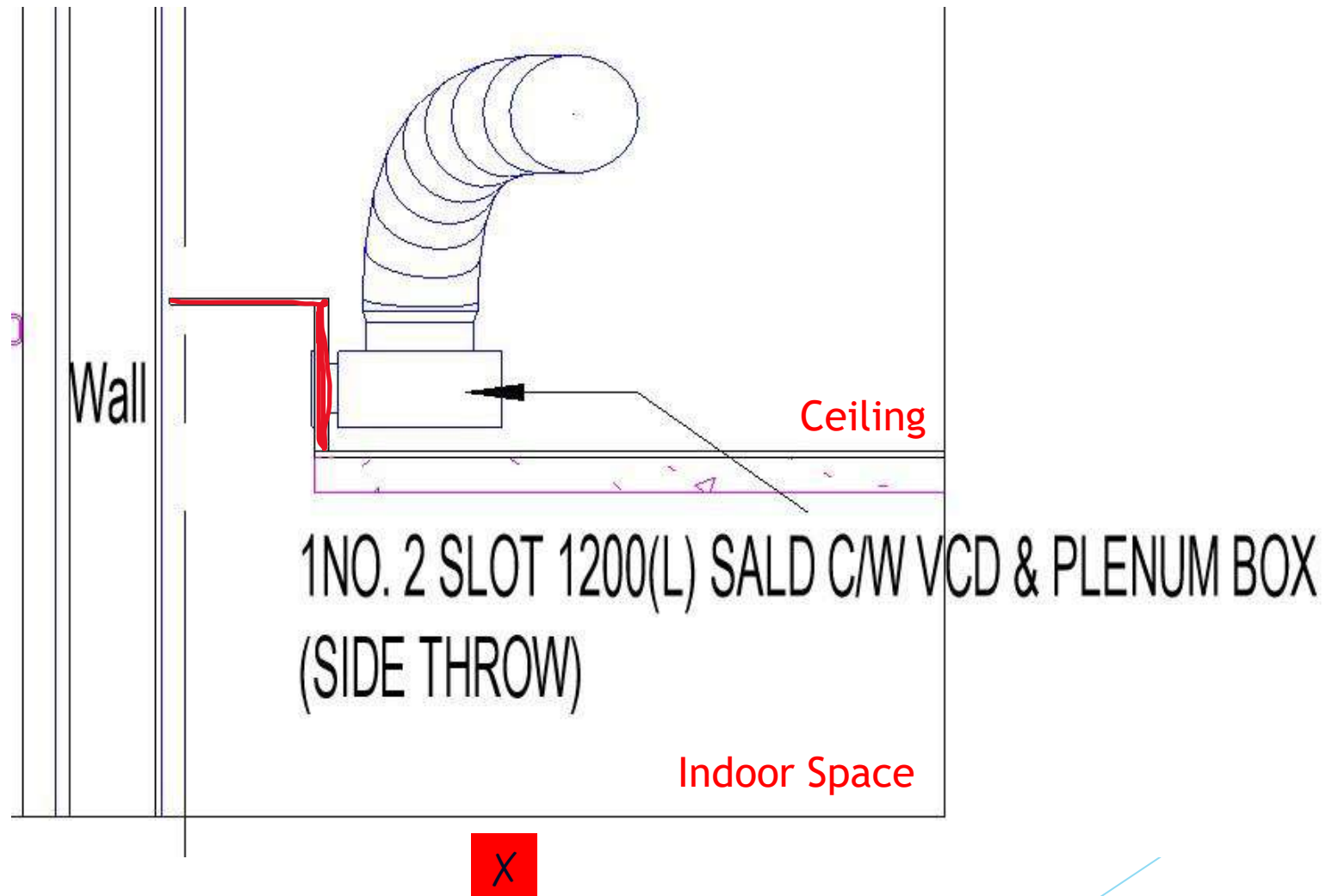


Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat

Design Requirements of Air Terminals

Section View of Linear Diffuser in Recess Cove

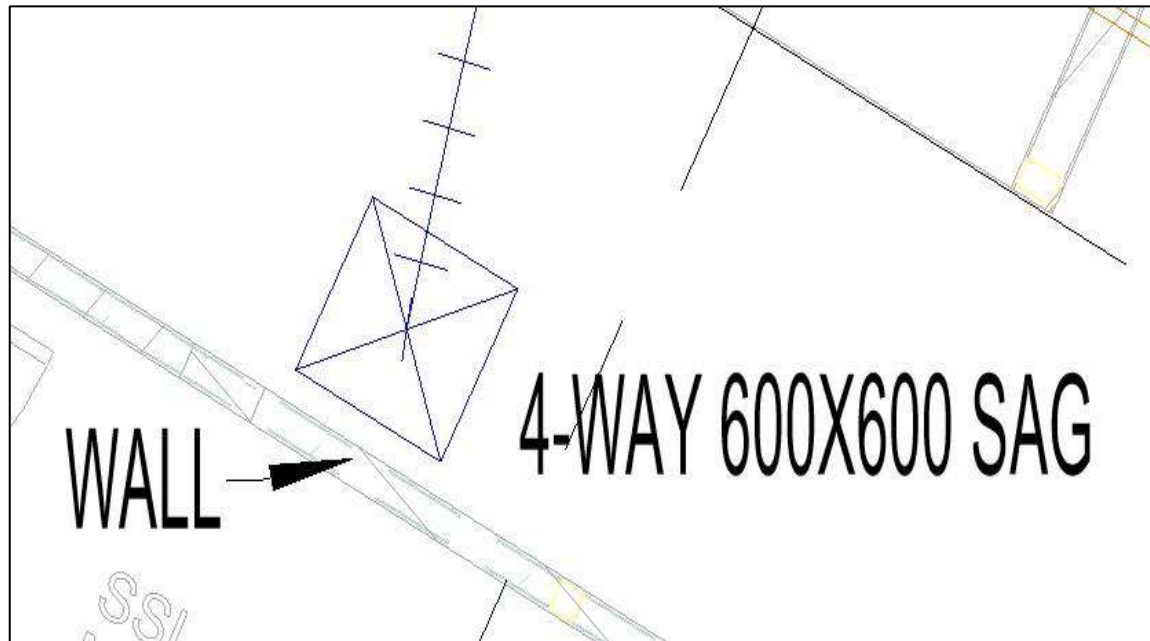


Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
2. Supply air should not be directed at walls

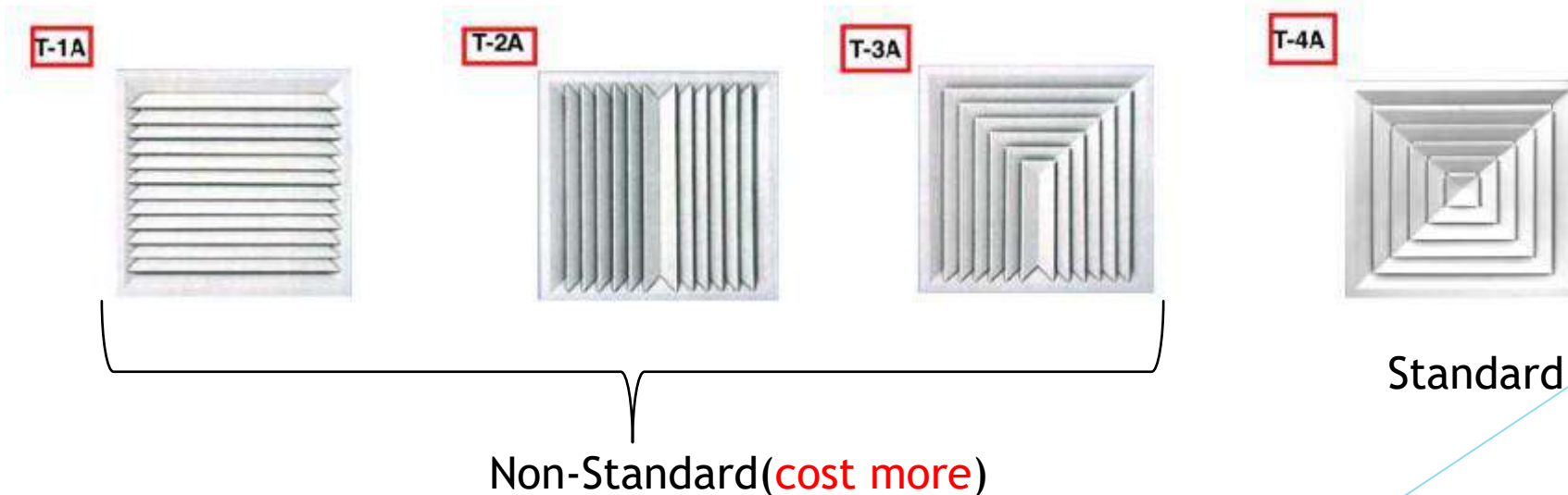
Design Requirements of Air Terminals

Plan Views of 4-Way Supply Diffuser



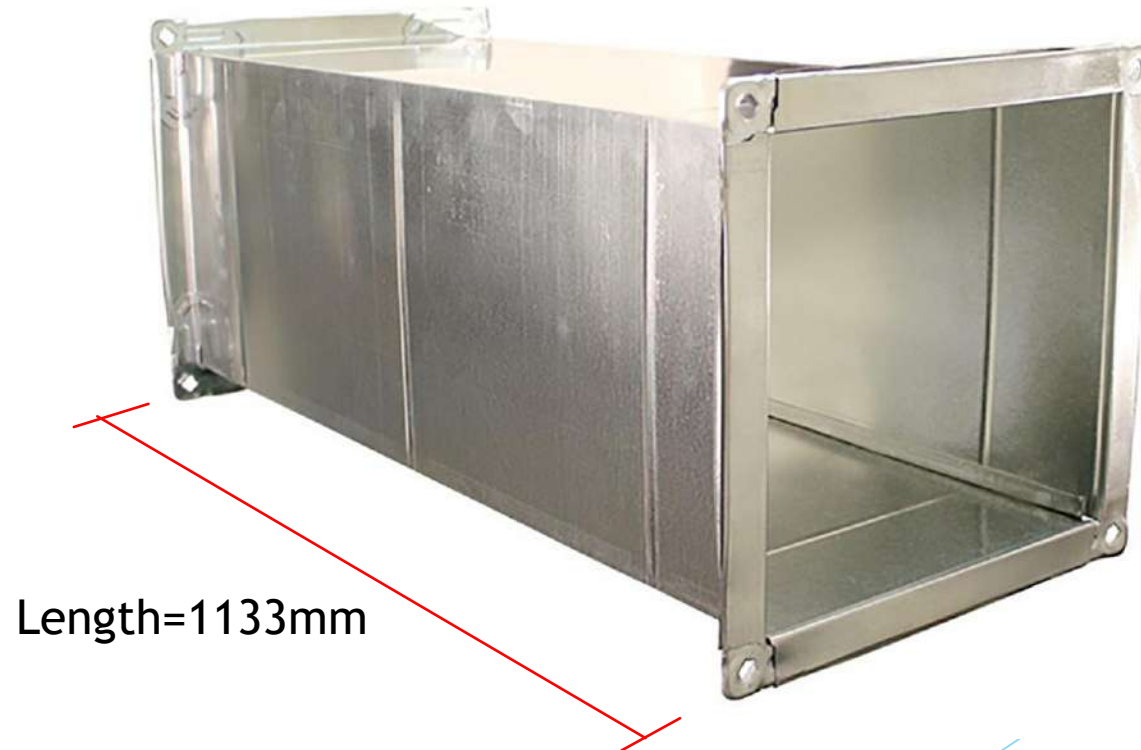
Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
 2. Supply air should not be directed at walls
 3. Should always use 4-way type diffusers if possible.
- (If situation don't allow, change the type of grille accordingly)



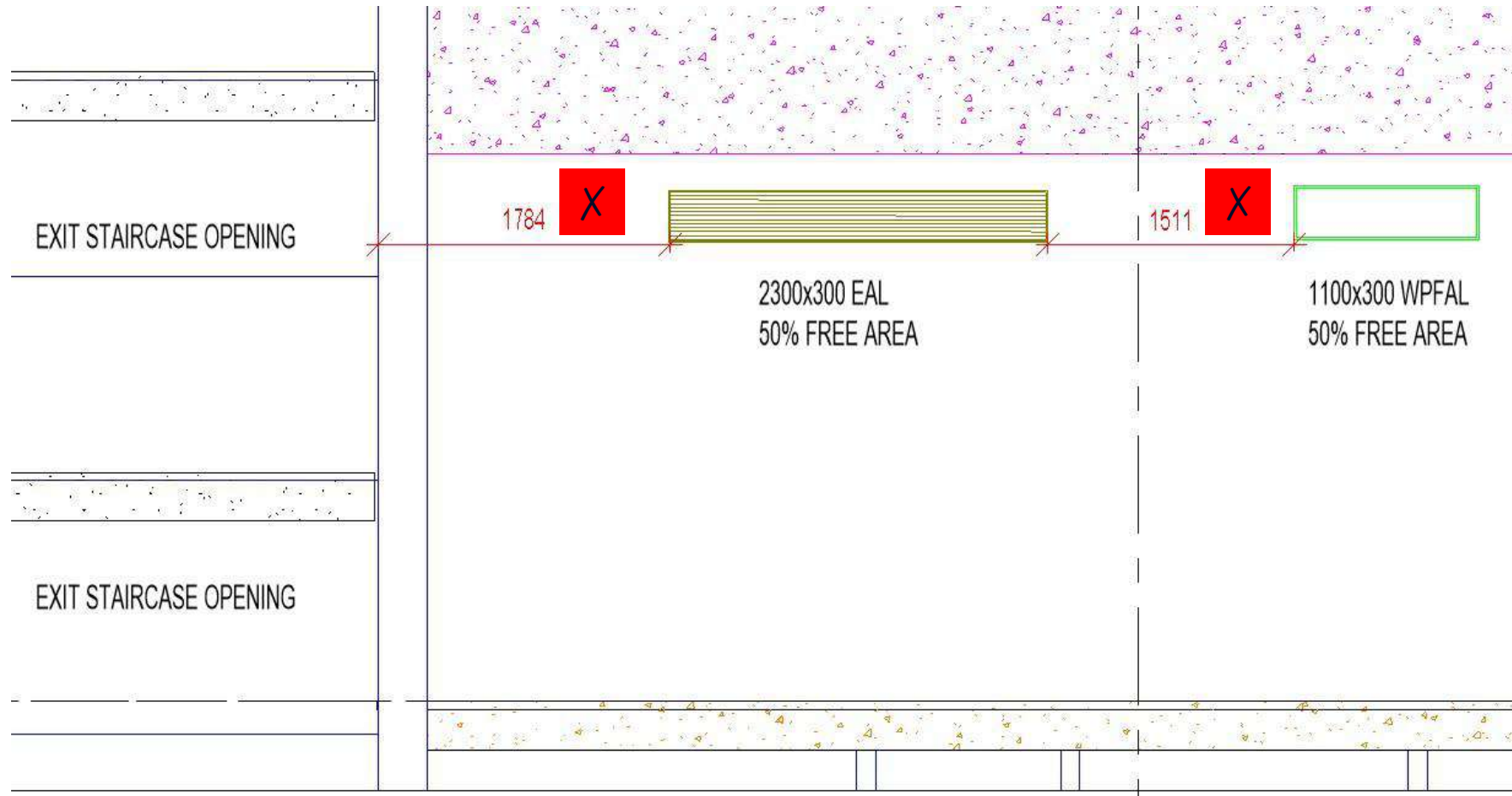
Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
2. Supply air should not be directed at walls
3. Number of directional openings should be considered during sizing
4. Length of Side-Throw Grille is to be $\leq 1000\text{mm}$



Design Requirements of Air Terminals

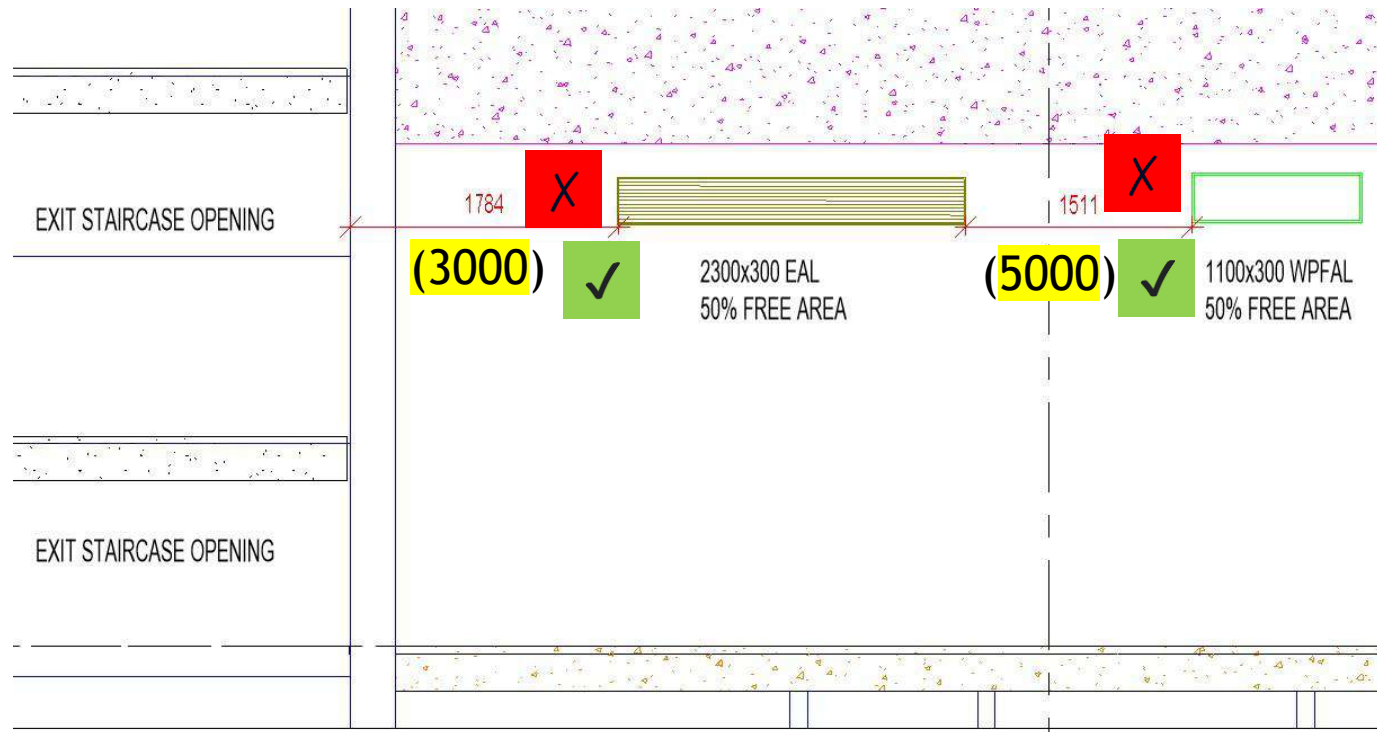
Section View of EAL and FAL



Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
2. Supply air should not be directed at walls
3. Number of directional openings should be considered during sizing
4. Length of Side-Throw Grille is to be $\leq 1000\text{mm}$
5. Ventilation Openings must be located at least 3m away from any exit staircase openings
6. Fresh air intake point must be at least 5m away from any Exhaust discharge openings

SCDF FIRE PRECAUTIONS
CODE OF PRACTICE
(2.3.2C) & (7.1.9B)



Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
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3. Number of directional openings should be considered during sizing
4. Length of Side Throw grille is to be $\leq 1000\text{mm}$
5. Ventilation Openings must be located more than 3m away from any exit staircase openings
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7. Double deflection grille is usually used for supply air

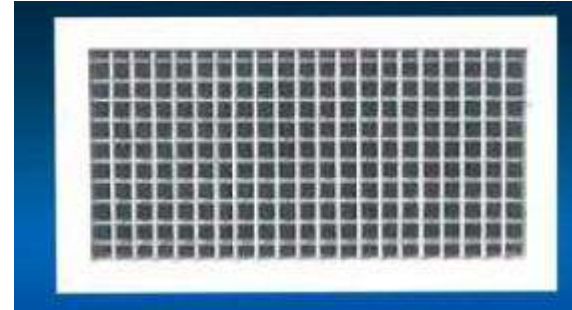
SCDF FIRE PRECAUTIONS
CODE OF PRACTICE
(7.1.9B)

Single Deflection Type



Horizontal/vertical blade

Double Deflection Type



Horizontal and vertical blade
(Cost more)

Design Requirements of Air Terminals

1. Supply air should not be directed at glass panes that are exposed to heat
2. Supply air should not be directed at walls
3. Number of directional openings should be considered during sizing
4. Allowable Width of Side Flush Direct-Duct mounted diffusers is to be $\leq 1100\text{mm}$
5. Ventilation Openings must be located more than 3m away from any exit staircase openings
6. Fresh air intake point must be more than 5m away from any exhaust discharge openings
7. Double deflection grille is usually used for supply air
8. Supply grilles/diffusers that are exposed to outdoor air to be coated with anti-condensation paint

SCDF FIRE PRECAUTIONS
CODE OF PRACTICE
(7.1.9B)

